

Data Science Capstone Project Report

SpaceX Falcon 9 First-Stage Landing Prediction

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GitHub repository containing all completed notebooks and Python files:
<https://github.com/razerblade09/spacex-falcon9-capstone>

This PDF is the completed capstone presentation, submitted as a PDF file, and it completes every required section: Executive Summary; Introduction; Data Collection by SpaceX API; Data Collection by web scraping; Data Wrangling Methodology; EDA with Data Visualization; EDA with SQL queries and results; Folium Map; Plotly Dash dashboard; Predictive Analysis classification, results and confusion matrix; and Conclusion with insights. All 19 finished slides follow this page, on pages 3 to 21, each carrying charts, tables, query output or written analysis rather than a heading alone.

Executive summary and introduction slides (slides 2 to 3, pages 4 to 5)

The executive summary slide states the methods and the key results together: 90 Falcon 9 launches from 2010 to November 2020 were collected, wrangled, explored with both visualization and SQL, mapped with Folium, and modelled with four classifiers, giving a 66.7 percent overall first-stage landing success rate and 83.3 percent test accuracy from the best model. The introduction slide sets out the problem: SpaceX advertises a Falcon 9 launch at 62 million dollars against 165 million or more from other providers, an advantage that exists only because the first stage is recovered and reused, so predicting whether that stage lands is what makes a competing bid costable.

Data collection and data wrangling methodology slides (slides 4 to 6, pages 6 to 8)

Data collection is shown twice. The API slide gives the request flow as a numbered pipeline: GET /v4/launches/past, filter to single-core single-payload Falcon 9 flights, then resolve each launch through the /rockets, /launchpads, /payloads and /cores endpoints into 90 one-row-per-launch records. The web scraping slide gives the parallel flow for the Wikipedia "List of Falcon 9 and Falcon Heavy launches" page using requests and BeautifulSoup, which both cross-checks the API data and supplies the SQL source table. The data wrangling slide documents the cleaning: PayloadMass was 5.6 percent missing and mean-imputed, LandingPad nulls were kept because a null legitimately means no landing was attempted, and the free-text Outcome field ("True ASDS", "False Ocean", "None None") was engineered into a binary Class label, giving 60 successful landings and 30 unsuccessful.

EDA with data visualization slides (slides 7 to 8, pages 9 to 10)

Six charts are presented and interpreted: scatter plots of flight number against launch site and against orbit type, scatter plots of payload mass against launch site and against orbit type, a bar chart of success rate by orbit, and a line chart of success rate by year. The analysis drawn from them is that heavier, higher-energy GTO and ISS missions land least reliably, and that success rate climbs from roughly 0 percent across 2010 to 2013 to close to 100 percent by 2019 and 2020, which makes landing success a maturing process rather than a fixed property of the hardware.

EDA with SQL slides and queries (slides 9 to 11, pages 11 to 13)

The scraped launch table was loaded into a SQL database as SPACEXTABLE and queried directly. The slides list all nine queries and then present their results: four distinct launch sites (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E); 45,596 kg of total payload flown for NASA CRS; 2,534.7 kg average payload for F9 v1.1 boosters; 1 May 2017 as the first successful ground-pad landing; 15,600 kg as the maximum payload carried; two failed drone-ship landings in 2015; the boosters that succeeded on a drone ship carrying 4,000 to 6,000 kg; and a full ranking of landing outcomes between 2010 and 2017, led by 20 successes against 10 no-attempts.

Folium map slides (slides 13 to 14, pages 15 to 16)

The Folium slides show all four launch sites plotted as markers together with every one of the 56 individual launch records, coloured green for a successful landing and red for a failure or no attempt, so the denser clusters are readable at a glance: KSC LC-39A landed 10 of 13 launches at 77 percent, while CCAFS LC-40, the earliest and highest-volume pad, landed only 7 of 26 at 27 percent. The proximity analysis slide measures distance from KSC LC-39A with the haversine formula to the nearest coastline at 7.43 km, the nearest city Titusville at 16.33 km, the nearest railway which is the NASA Railroad at 0.69 km, and the nearest highway Kennedy Parkway North at 0.85 km, with the rail and road coordinates taken from OpenStreetMap.

Plotly Dash dashboard slides (slides 12, 15 and 16, pages 14, 17 and 18)

The dashboard slides show the built Plotly Dash application and its outputs: a launch-site dropdown driving a live success-rate pie chart, presented both as the share of all successful landings contributed by each site and as the success-versus-failure split within the selected site, plus a payload-mass range slider over a payload-versus-outcome scatter plot coloured by booster version category. The reading given is that newer booster generations (FT, B4, B5) succeed across a far wider payload range than the early v1.0 and v1.1 boosters.

Predictive analysis, classification and results slides (slides 17 to 18, pages 19 to 20)

The model development slide gives the full setup: 80 one-hot-encoded feature columns plus flight number, payload mass, flights, grid fins, reused, legs and block, standardised with StandardScaler, split 80/20 into 72 training and 18 test launches, with Logistic Regression, Support Vector Machine, Decision Tree and K-Nearest Neighbors each tuned by 10-fold GridSearchCV over a stated hyperparameter grid. The results slide gives the comparison table and the confusion matrix: Logistic Regression, SVM and KNN all reach 83.3 percent test accuracy and the Decision Tree 77.8 percent, and on the 18 test launches the best model returns 12 true positives, 3 true negatives, 3 false positives and no false negatives, so it never misses a booster that actually lands.

Conclusion slide, creativity and innovative insights (slide 19, page 21)

The conclusion slide names Logistic Regression as the best model with its tuned parameters ($C=1$, L2 penalty, lbfgs solver) and justifies the choice: it ties SVM and KNN on accuracy but is the only one whose coefficients explain a cost bid to a non-technical stakeholder, and it is the cheapest to retrain as launches accumulate. The insights it closes on are that landing success is highly learnable from public launch parameters, that orbit type and payload mass are the strongest predictors, and that the year-over-year improvement means a competing bidder should treat the model as a first-pass cost-risk estimator and retrain it continuously rather than trusting a fixed success rate. The deck itself is built programmatically from a script in the repository, and its Folium and dashboard figures are rendered over live OpenStreetMap tiles rather than reproduced from the lab.

SpaceX Falcon 9 First-Stage Landing Prediction

Data Science Capstone Project Report

Executive Summary

Methods used: Data was collected via the SpaceX REST API and Wikipedia web scraping, cleaned and labeled, explored with visualization and SQL, mapped geospatially with Folium, and modeled with four classifiers (Logistic Regression, SVM, Decision Tree, KNN) tuned via GridSearchCV.

Key results: Across 90 Falcon 9 launches, the overall first-stage landing success rate was 67%. The best-performing model, Logistic Regression, reached 83% test accuracy in predicting landing outcome.

90

Launches analyzed

67%

Overall landing success rate

83.3%

Best model test accuracy

4

ML models compared

Conclusion: Landing success is highly predictable from launch parameters, meaning a competing launch provider could reliably estimate SpaceX's true reusable-launch cost advantage.

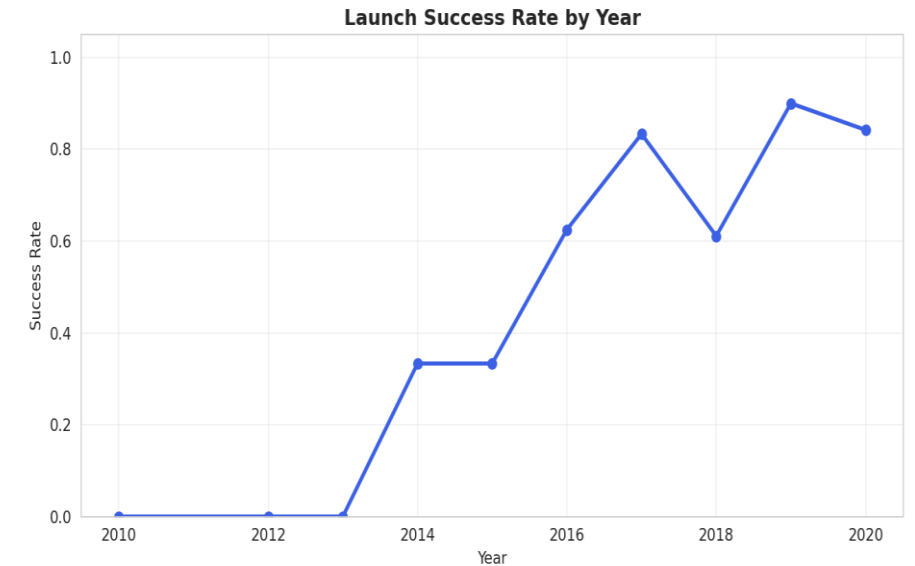
Introduction

SpaceX advertises Falcon 9 launches at \$62 million, versus \$165M+ from other providers — largely because SpaceX reuses the rocket's first stage instead of discarding it.

Problem statement: If the first stage does not land successfully, that launch's cost estimate changes substantially. This project builds a model to predict whether the first stage will land, using historical launch data — enabling accurate cost estimation for a competing bid.

Objectives:

- Collect and clean historical Falcon 9 launch data
- Identify factors correlated with landing success
- Visualize patterns across sites, orbits, and time
- Build and evaluate a landing-outcome classifier



Falcon 9 landing success climbed steadily as SpaceX matured its recovery process.

Data Collection – SpaceX API

Pipeline:

1

GET /v4/launches/past → all historical launches

2

Filter to single-core, single-payload Falcon 9 flights

3

For each launch, call /rockets, /launchpads, /payloads, /cores endpoints

4

Assemble into one row per launch: booster, orbit, site, outcome, mass

```
api.spacexdata.com/v4/launches/past
```

Result: 90 clean Falcon 9 launch records (2010–Nov 2020), each with booster version, payload mass, orbit, launch site, flight count, grid fins, reused flag, legs, landing pad, and outcome.

Falcon 1 launches were excluded — this project focuses only on the Falcon 9 landing-prediction problem.

Data Collection – Web Scraping

Pipeline:

- 1 Request the Wikipedia "List of Falcon 9 and Falcon Heavy launches" page
- 2 Parse HTML tables with BeautifulSoup, locate the wikitable elements
- 3 Extract column headers, then walk each row of launch data
- 4 Clean text (dates, booster serials, payload mass) into a dataframe

```
BeautifulSoup(response.text, 'html.parser')
```

This gives an independent, cross-checked source for launch records — used to validate the API dataset and fill in fields (like precise payload customer) the API doesn't expose.

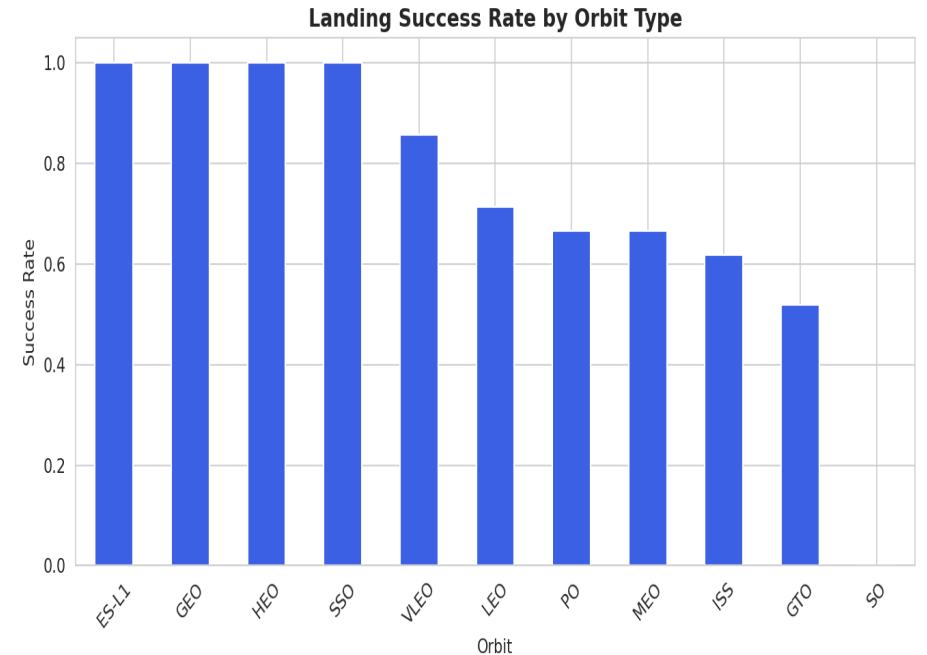
Key techniques: regex/Unicode cleanup for payload mass strings, and helper functions to pull booster version and landing status out of merged table cells.

Data Wrangling Methodology

Missing values: PayloadMass (5.6% missing) filled with column mean; LandingPad (28.9% missing) kept as-is since a null pad legitimately means "no ground/drone-ship landing occurred."

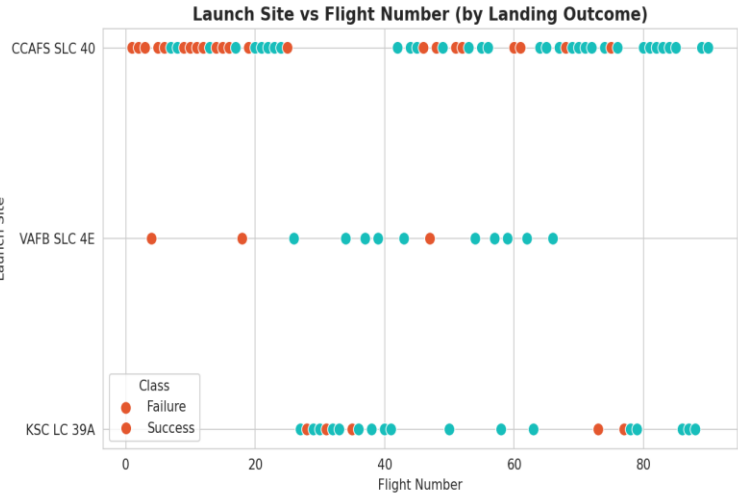
Label engineering: The raw Outcome field (e.g. "True ASDS", "False Ocean", "None None") was converted into a binary Class label — 1 if the booster landed successfully by any method, 0 otherwise.

Result: 90 launches → 60 successful landings (66.7%), 30 unsuccessful (33.3%)

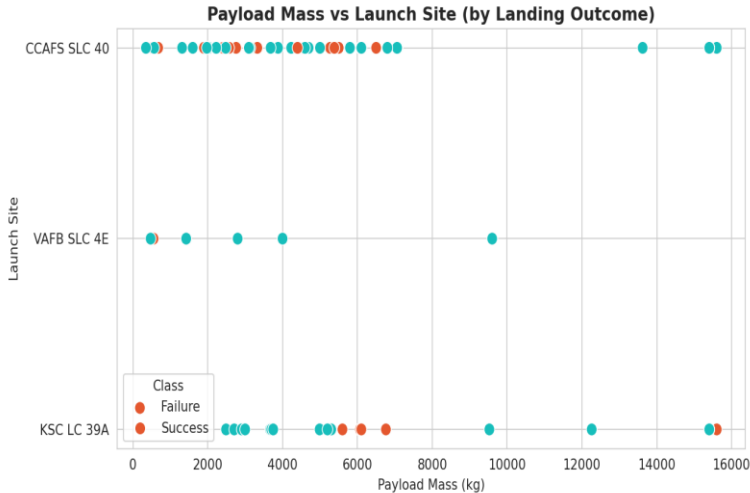


EDA with Data Visualization

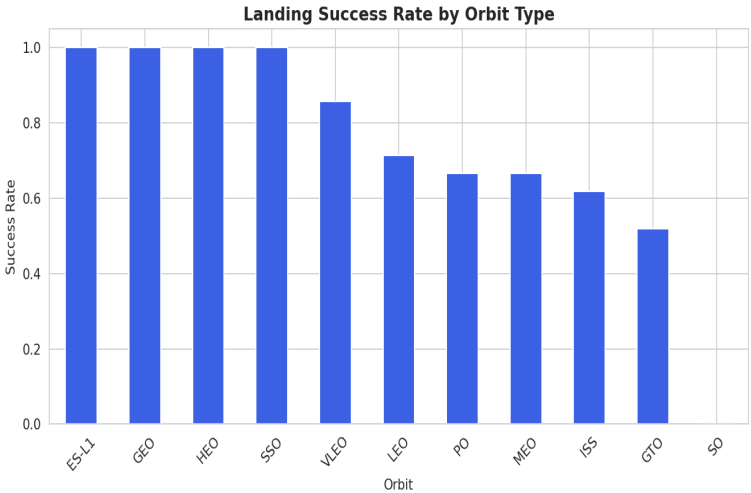
We plotted flight number, payload mass, orbit type, and launch year against landing outcome to surface which factors correlate with success.



Flight # vs Launch Site



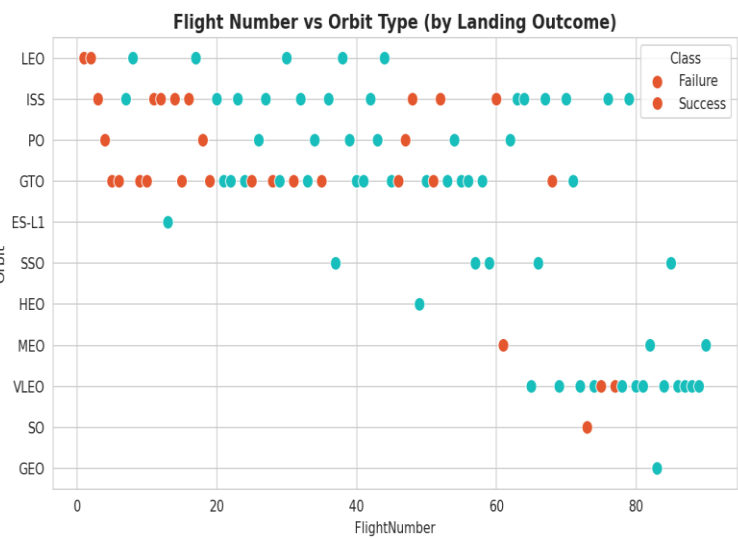
Payload vs Launch Site



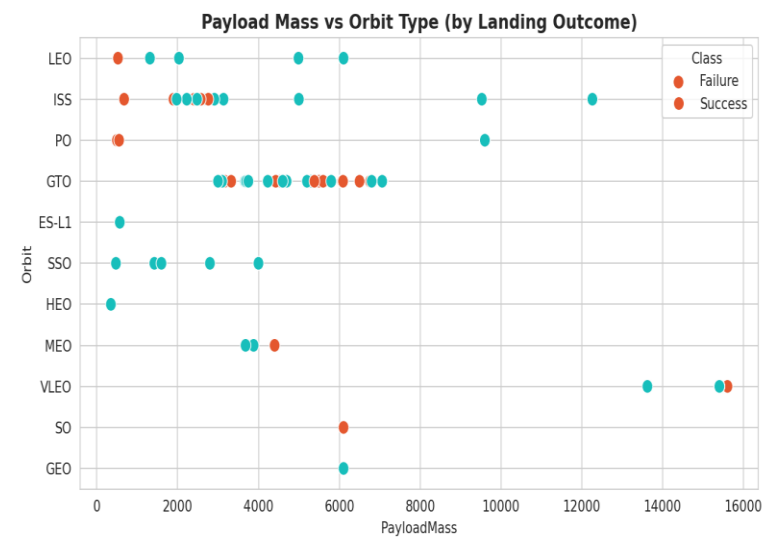
Success Rate by Orbit

Chart types used: scatter plots (flight #/payload vs site or orbit), bar charts (success rate by orbit), and a line chart (yearly trend, see Introduction slide).

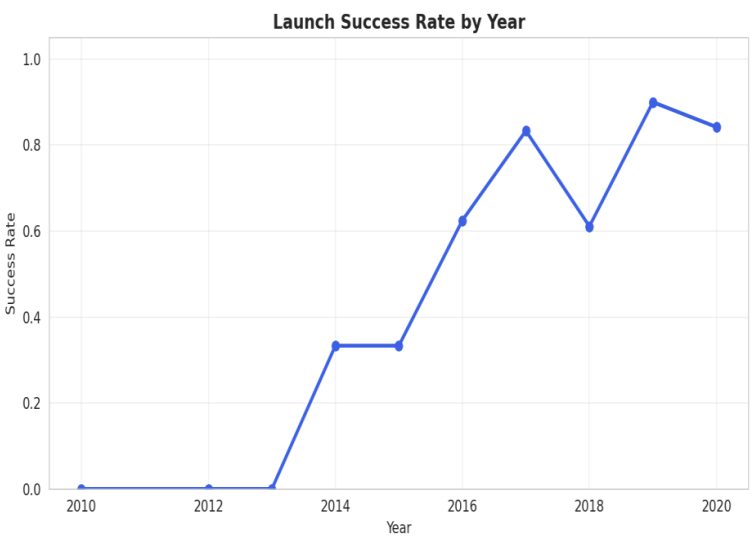
EDA Visualization — Orbit & Trend Detail



Flight # vs Orbit Type



Payload vs Orbit Type



Success Rate by Year

Takeaway: GTO and ISS-orbit missions (heavier, higher-energy) show lower success rates; success climbed year-over-year as SpaceX refined the recovery process, from ~0% in 2010–2013 to consistently near 100% by 2019–2020.

EDA with SQL

The scraped launch table was loaded into a SQL database (SQLite / IBM Db2) as SPACEXTABLE, then queried directly to answer specific business questions.

- ✓ Distinct launch sites used
- ✓ Total payload mass carried for NASA (CRS)
- ✓ Average payload mass for F9 v1.1 boosters
- ✓ Date of the first successful ground-pad landing
- ✓ Boosters that succeeded on a drone ship with 4,000–6,000 kg payload
- ✓ Count of each mission outcome (success/failure)
- ✓ Booster(s) carrying the maximum payload mass
- ✓ Failed drone-ship landings in 2015
- ✓ Ranking of landing outcomes between 2010–2017

EDA SQL Results — Sites & Payload

Launch Sites (4 distinct)

CCAFS LC-40 · CCAFS SLC-40 · KSC LC-39A · VAFB SLC-4E

Total payload for NASA (CRS): 45,596 kg

Avg. payload, F9 v1.1 boosters: 2,534.7 kg

Max payload carried: 15,600 kg (F9 B5)

Milestones

First successful ground-pad landing: 1 May 2017

2015 drone-ship landing failures: 2 (F9 v1.1 B1012, B1015 — CCAFS LC-40)

4,000–6,000 kg drone-ship successes: F9 FT B1021.2, B1022, B1026, B1031.2

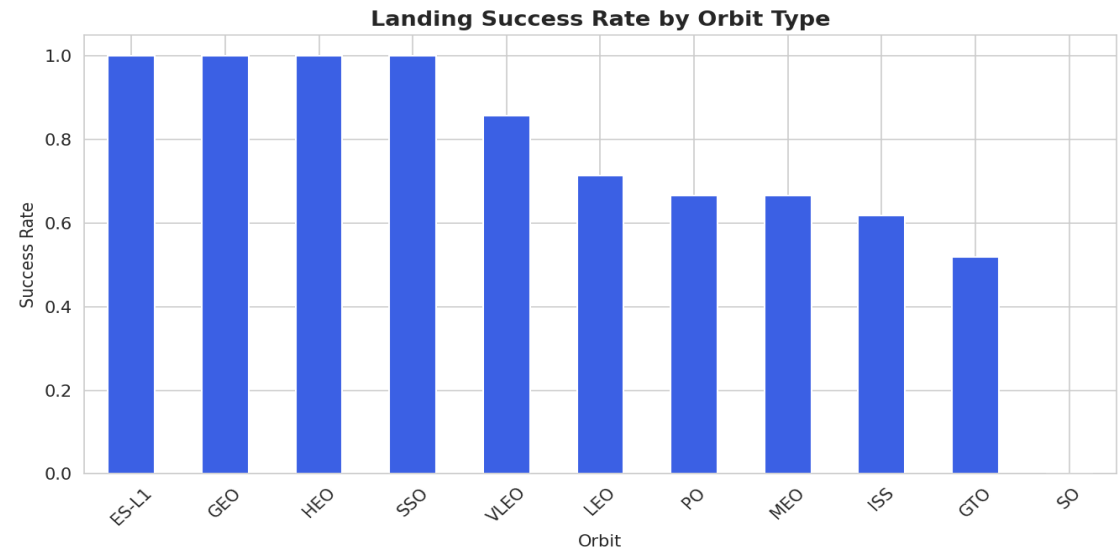
These figures ground the EDA charts in exact, query-verified numbers rather than visual estimates.

EDA SQL Results — Success Rates & Rankings

Landing outcome ranking, 2010–2017 (by count):

Landing Outcome	Count
Success	20
No attempt	10
Success (drone ship)	8
Success (ground pad)	6
Failure (drone ship)	4
Failure	3
Controlled (ocean)	3
Failure (parachute)	2

Mission outcome (2010–2020): 98 of 101 recorded outcomes were successful missions — first-stage recovery success is the harder, separate problem this model targets.

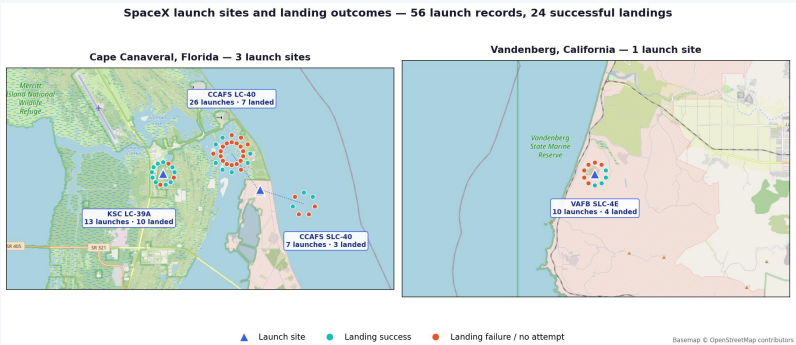


Interactive Visual Analytics

Two interactive tools let stakeholders explore the data themselves rather than reading static charts:

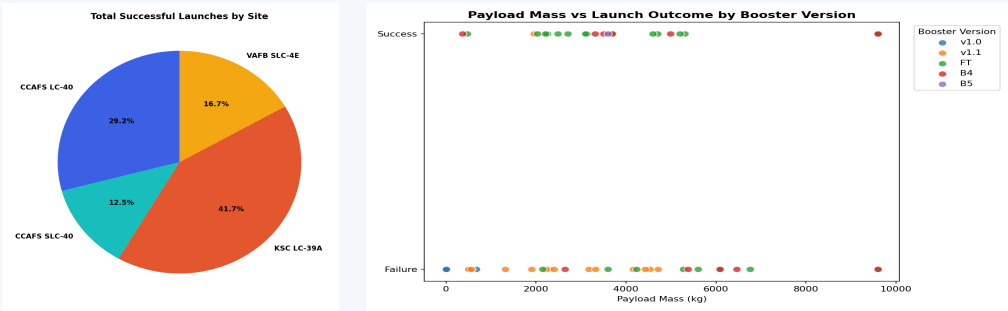
Folium Map

Launch-site markers, per-launch outcome markers (color-coded), and proximity distance lines (coastline, city, railway, highway).



Plotly Dash App

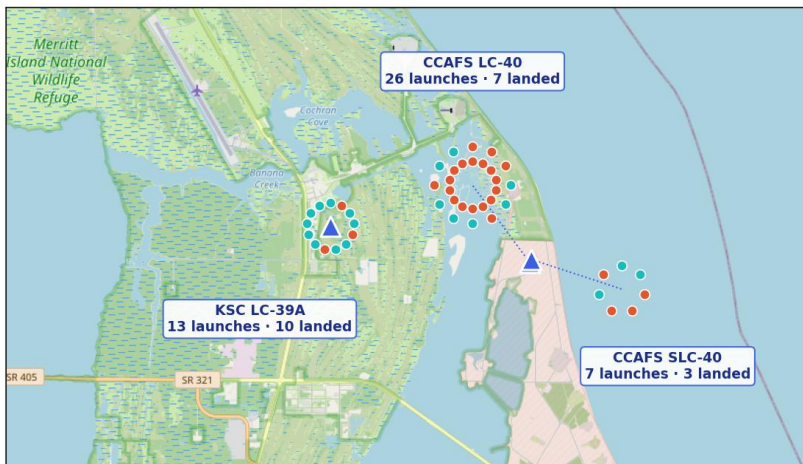
Dropdown to filter by launch site; live-updating success-rate pie chart and payload-vs-outcome scatter plot.



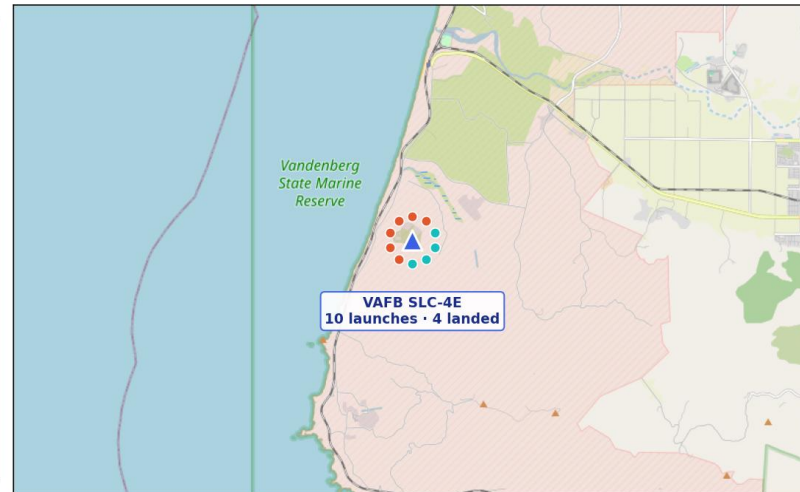
Launch Site Markers & Launch Records

SpaceX launch sites and landing outcomes — 56 launch records, 24 successful landings

Cape Canaveral, Florida — 3 launch sites



Vandenberg, California — 1 launch site



▲ Launch site ● Landing success ● Landing failure / no attempt

Basemap © OpenStreetMap contributors

All 4 launch sites plotted with markers.

Every individual launch (56 records) plotted as a marker cluster, colored green for a successful landing and red for a failed/no-attempt landing — letting you see which sites cluster more successes.

Observation: KSC LC-39A shows the densest green cluster — 10 landings from 13 launches (77%). CCAFS LC-40, the earliest and highest-volume pad, landed only 7 of 26 (27%): most of its records predate a working recovery process.

Proximity Analysis

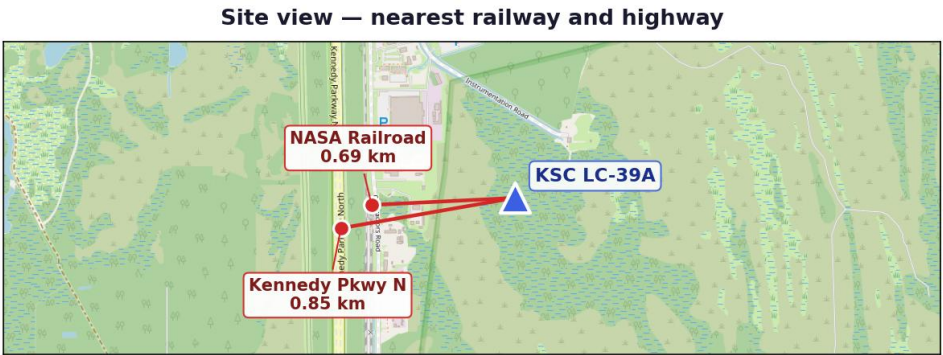
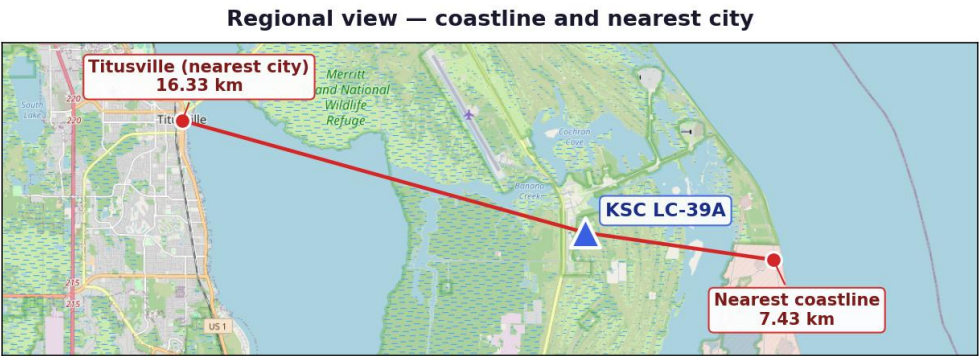
Using the haversine formula, we measured distance from KSC LC-39A to nearby geographic features:

Feature	Distance from Launch Site
Nearest coastline point	7.43 km
Nearest city (Titusville)	16.33 km
Nearest railway (NASA Railroad)	0.69 km
Nearest highway (Kennedy Pkwy N)	0.85 km

Why this matters: Launch sites are deliberately sited close to coastlines (for downrange safety over open water) and to rail and road links (boosters and stages arrive overland), but a safe distance from population centers — real siting constraints any competing launch provider would need to replicate.

City, railway and highway coordinates sourced from OpenStreetMap.

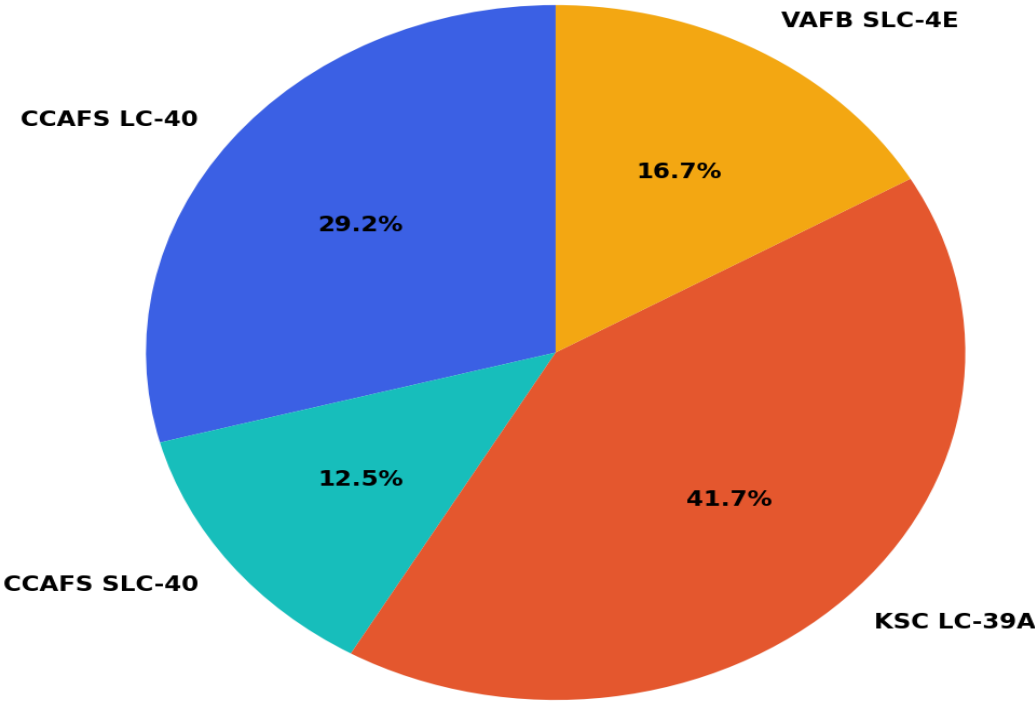
Proximity analysis — distances from KSC LC-39A



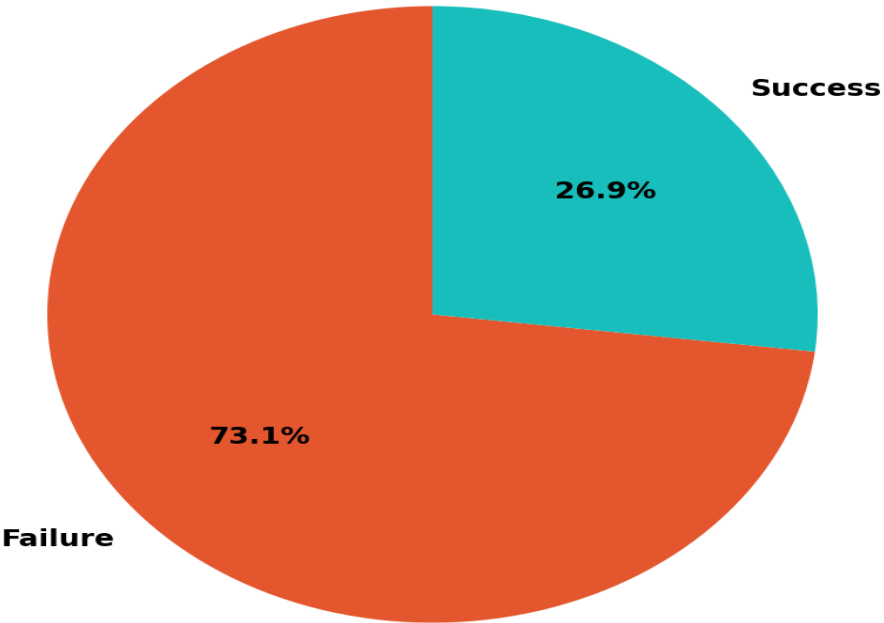
Basemap © OpenStreetMap contributors

Success Rate — Pie Charts

Total Successful Launches by Site

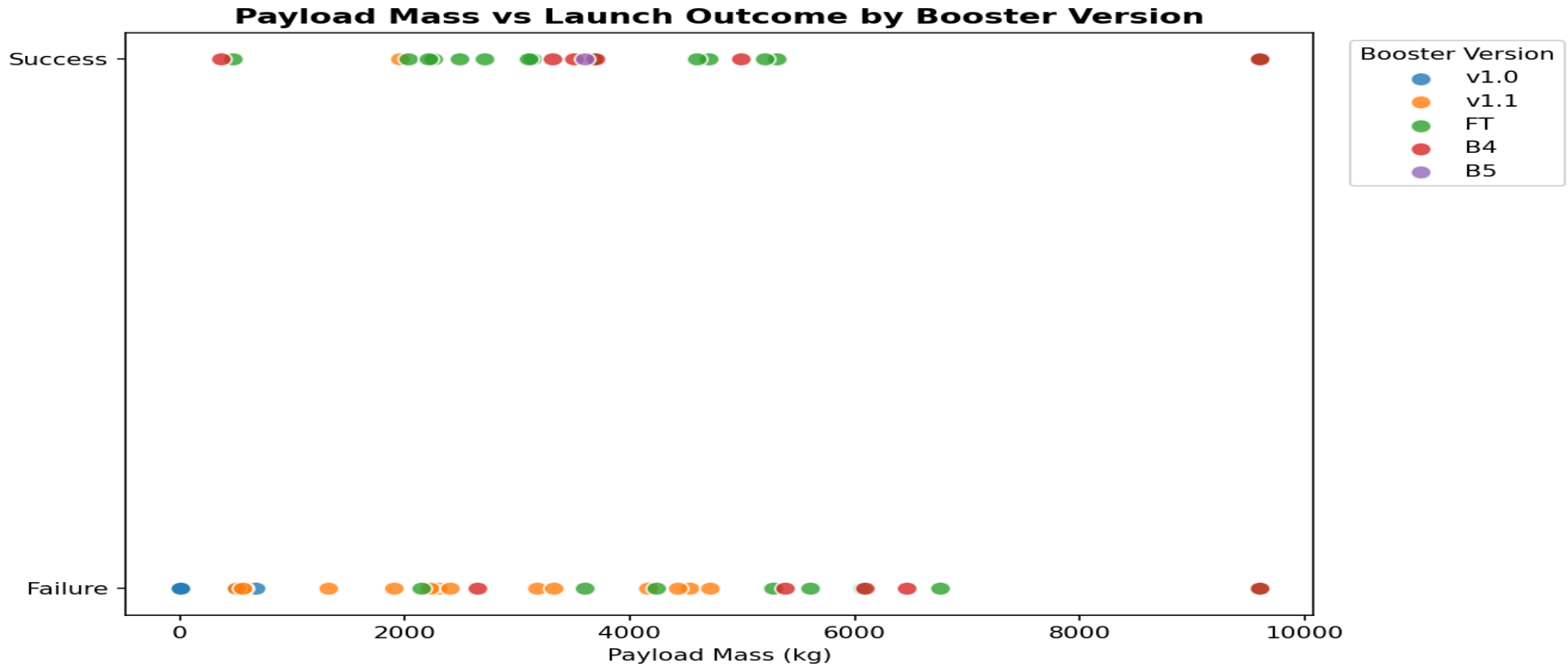


Success vs Failure - CCAFS LC-40



Left: share of total successful landings contributed by each site. Right: success-vs-failure split at the top-volume site (CCAFS LC-40) — filterable live via the dashboard dropdown.

Payload Mass vs. Launch Outcome



A RangeSlider (0–10,000 kg) lets users narrow the payload window; the scatter plot recolors by booster version category (v1.0 / v1.1 / FT / B4 / B5), revealing that newer booster generations succeed reliably across a much wider payload range than early versions.

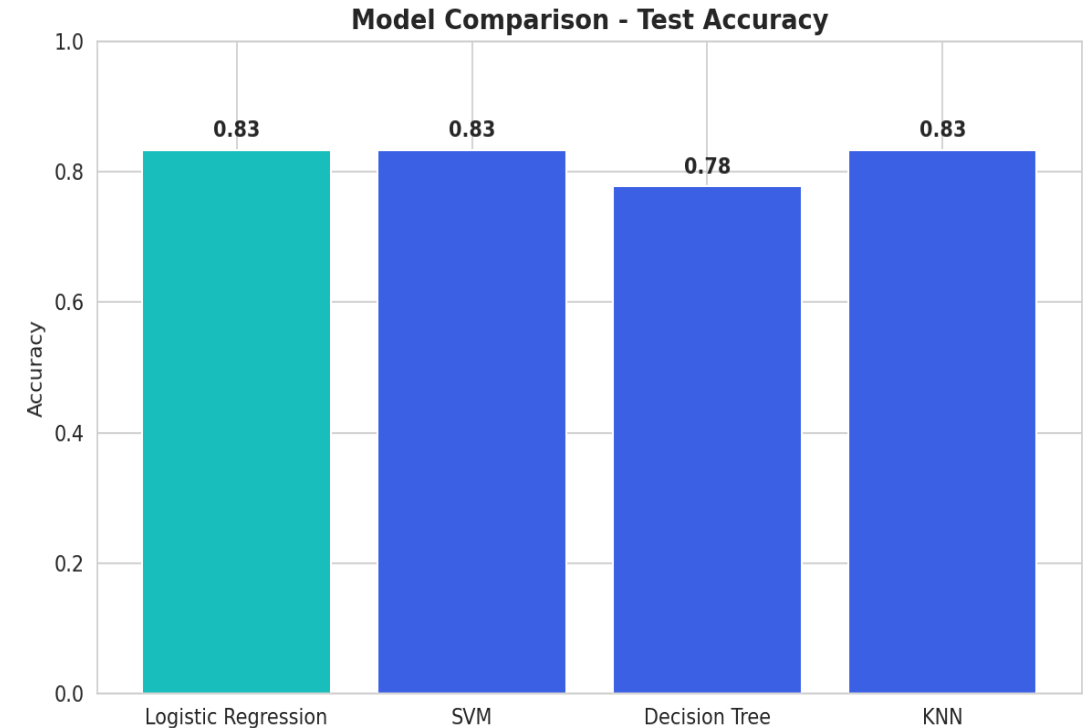
Model Development

Features: 80 one-hot-encoded columns (orbit, launch site, landing pad, booster serial) plus flight number, payload mass, flights, grid fins, reused, legs, and block — standardized with StandardScaler.

Split: 80% train / 20% test (72 / 18 launches), stratified via `random_state=2`.

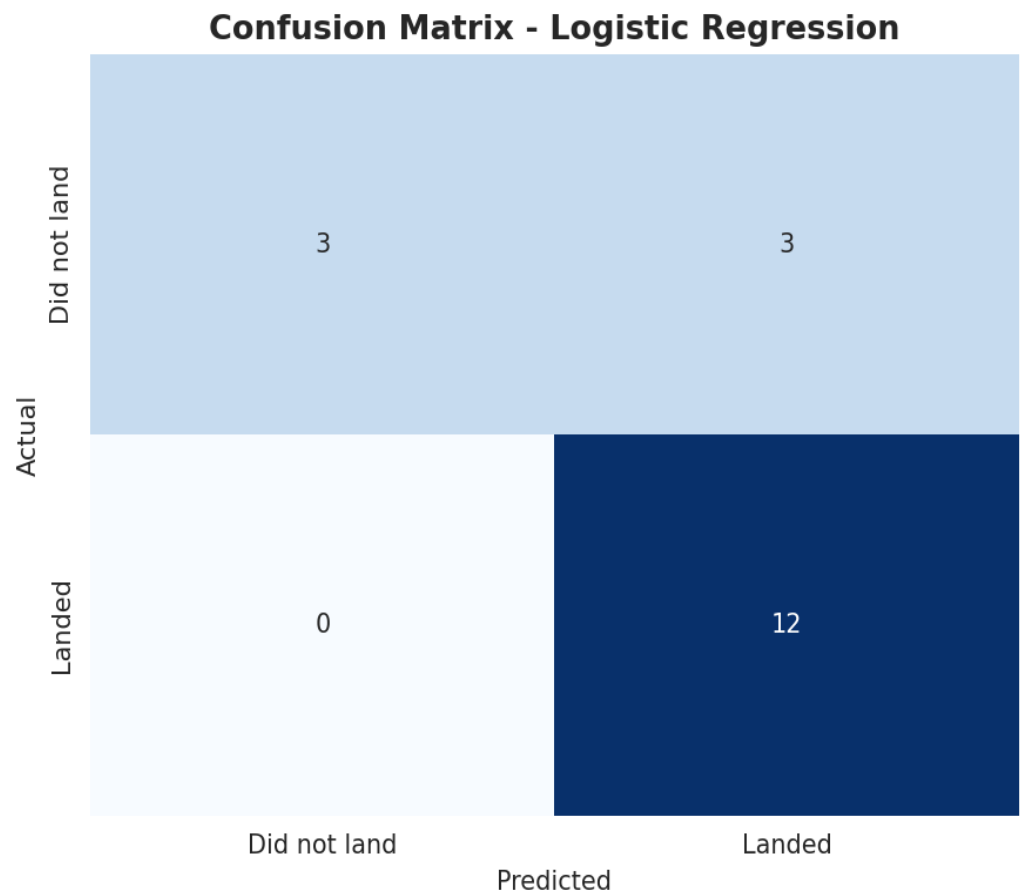
Models compared (each tuned via 10-fold GridSearchCV):

- Logistic Regression — C, penalty, solver
- Support Vector Machine — kernel, C, gamma
- Decision Tree — criterion, max_depth, min_samples_split
- K-Nearest Neighbors — n_neighbors, distance metric (p)



All four models converged near 83% test accuracy, with Decision Tree slightly behind — signal that the engineered features are strong and model choice matters less than feature quality here.

Evaluation Results & Confusion Matrix



Model	CV Score	Test Accuracy
Logistic Regression	0.821	0.833
SVM	0.834	0.833
KNN	0.834	0.833
Decision Tree	0.804	0.778

Reading the confusion matrix (18 test launches): 12 successful landings correctly predicted, 3 unsuccessful landings correctly predicted, 3 unsuccessful landings misclassified as successful, 0 successful landings missed. The model never misses a true landing — it's conservative in the safer direction for cost bidding.

Best Model & Conclusion

Best Model: Logistic Regression

Best params: C=1, L2 penalty, lbfgs solver

Why it wins: Ties SVM and KNN on raw accuracy (83.3%) but is far more interpretable — each feature's coefficient directly shows its effect on landing-success odds, which matters for explaining a cost bid to stakeholders. It's also the cheapest to retrain as new launches arrive.

Conclusion & Insights

- Landing success is highly learnable from launch parameters — 83% test accuracy from a simple, interpretable model.
- Orbit type and payload mass are the strongest visual predictors; heavier GTO missions land less reliably than light LEO/ISS missions.
- Success rate improved dramatically over time (2013 → 2020), reflecting SpaceX's iterative process improvements — not a static property of the hardware.
- Recommendation: a competing bidder can use this model as a first-pass cost-risk estimator, refining it further with more recent launches and continuous re-training.